Reg. No.										

MANIPAL INSTITUTE OF TECHNOLOGY

MANIPAL

(A constituent unit of MAHE, Manipal)

V SEMESTER B.TECH. (MECHANICAL / I&P ENGINEERING)

END SEMESTER Make-Up EXAMINATIONS, DECEMBER 2019

SUBJECT: CAD-CAM [MME 3103]

REVISED CREDIT SYSTEM

Time: 3 Hours MAX. MARKS: 50

Instructions to Candidates:

- Answer **ALL** the questions.
- Missing data may be suitably assumed.
- Draw the sketches using PENCIL only

1A With a block diagram explain how the mechanical components were getting designed using Shigley's conventional design process. **4M**

1B Using the recursive relation, evaluate the points on the ellipse for one quadrant with an angle increment of 30°. The center of the ellipse is at (16, 24) with major and minor axis length being 24 and 16 units respectively. **3M**

1C What are work cell control and interlocks **3M**

2A Derive the equation of tangents at the first and last points for Bezier curve defined by four control points. **4M**

2B How 3D modelling with booleon operation differs from 3D modelling using boundary representation **3M**

2C Explain three network configurations used in CAD-CAM system **3M**

3A Derive an expression for the position vector of a hermite cubic spline curve in the parametric form by applying appropriate boundary conditions **5M**

3B A ruled surface is defined by two Bezier curves. One curve has control points $[2\ 9\ 5]^T$, $[5\ 6\ 1]^T$, $[9\ 3\ 3]^T$. The other curve has control points $[3\ 5\ 6]^T$, $[5\ 4\ 8]^T$, $[9\ 2\ 9]^T$ and $[11\ 1\ 10]^T$. Assume the origin of the ruled surface parameters at the lower left corner of the surface, compute the coordinates of the point on the surface at v=0.25 and u=0.50. **5M**

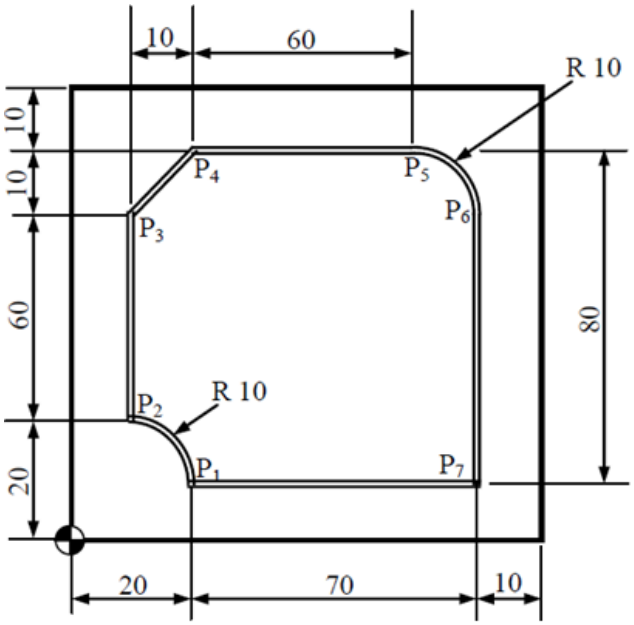
4A Explain any 4 specifications that is helpful to design a robot **4M**

4B Sketch the work volume of any 3 robot configurations **3M**

4C A scaling factor of 2.5 is applied in the Y direction while no scaling factor is applied in the X direction to a line whose end points coordinates are (2, 4) and (5, 9). Then the line is to be rotated subsequently through an angle of 40°. All the transformation operations takes place about the origin. Determine the necessary transformation matrix for these sequential operations and find the new coordinates of the end points of the line. **3M**

5A With examples explain how Adaptive Control Optimization and Adaptive Control Constraints aid in improving the productivity **4M**

5B Write a part program to perform the contour slotting operation on the component as shown in the Figure 5B. Depth of the contour = 1mm.



Billet Size : 100 x 100 x 10 mm

Cutter Dia: 8 mm

Figure 5B. **3M**

5C Explain any 3 types of FMS layouts with the schematic diagrams **3M**